## Comparative Table

| Tenecteplase | Global ID_Version<br>206893_11771375_1.0 |
|---|---|
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A consolidated 3.2.S.2.6 document is provided, compiling previous documents into a single document. It contains an overview of the complete manufacturing process history including the respective comparability studies in a concise and state-of-the-art format.

The following previous manufacturing process development documents are combined:

- Comparability study between Genentech and Boehringer Ingelheim: May 2013 (7776-s260bioal0101 – A144176)
- Data from a modified nanofiltration step: July 2014 (7776-s260abioal0401 – q00216834-01)
- Data from a modified Mn spike step: August 2018 (7776-s260abioal0601 – q00259098-01)
- Data from a adapted Phenyl Sepharose hydrophobic interaction chromatography step: April 2025 (206893-10970210_1.0)

The following previous manufacturing process development documents are not included:

- May 2013 (7776-s260bioal0201 – A144178)
- May 2013 (7776-s260bioal0301 – A141255)
- August 2018 (7776-s260bioal0501 – q00258489-01)

These documents were supportive documentation at the point in time of conducting the studies and are not considered relevant anymore. The first two documents (May 2013) summarize the revalidation of the Virosolve 180kD process step (virus removal). As the Virosolve 180 kDa membrane is replaced with a Planova BioEX the document (July 2014, 7776-s260abioal0401 – q00216834-01) are not relevant anymore. The relevant data are included in the combined document. The third document (August 2018, 7776-s260bioal0501 – q00258489-01) describes the rational of a slight modification of the manufacturing process step: addition of Manganese to the cell culture process. This data are not considered relevant anymore as the updated process is fully adapted. The relevant comparability data are included in the combined document.

The clarity and structure of analytical comparability information from the initial PPQ up to date was enhanced to align with current regulatory expectations and provide a clearer presentation of information.

Furthermore, data from process knowledge improvement (PKI) studies were included. These PKI studies were performed 2023 as part of continuous process verification (CPV) in order to

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improve process knowledge and process understanding. These studies include new analytical data, historical manufacturing knowledge, statistical methodologies, risk assessment, scale-down model and small-scale studies.
As a result of the PKI investigations, the process control strategy is refined by implementation of a state-of-the-art presentation of critical quality attributes (CQAs), process parameter classification (CPPs, KPPs, etc.) and definition of NORs and PARs for process parameters, in line with current regulatory requirements.

This also includes justification for the ranges of process parameters used in the commercial manufacturing process, as outlined in section 3.2.S.2.2.

Table 1 lists the changes between the currently approved and the proposed Manufacturing Process Development section 3.2.S.2.6.

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Table 1 Comparative table for section 3.2.S.2.6 Manufacturing Process Development

| **Current documents**<br>**7776-s260bioal0101 – A144176**<br>**7776-s260abioal0401 – q00216834-01**<br>**7776-s260abioal0601 – q00259098-01**<br>**206893-10970210_1.0** | **Proposed document**<br>**206893_11135038_1.0** | **Comments/ Justification for change** |
|---|---|---|
| N/A | **1. Introduction** | Provides and introduction together with reviewer guidance indicating what information is contained in each chapter. |
| N/A | **2. Tenecteplase drug substance critical quality attributes** | Describes the assessment of the critical quality attributes (CQA). |
| N/A | **3. Manufacturing process history** | Describes an overview of the drug substance manufacturing process development history. |
| N/A | **4. Drug Substance manufacturing process knowledge improvement and characterization activities** | Describes the risk assessment, process knowledge improvement program and process characterization studies. |
| N/A | **5. Summary of overall control strategy for commercial manufacturing of drug substance** | Provides a summary of the refined control strategy of the commercial drug substance manufacturing process. |
| **N/A** | **6. Process comparability studies**<br>**6.1 Overview of analytical comparability studies** | Provides an overview of the comparability studies conducted |

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| **Current documents**<br>**7776-s260bioal0101 – A144176**<br>**7776-s260abioal0401 – q00216834-01**<br>**7776-s260abioal0601 – q00259098-01**<br>**206893-10970210_1.0** | **Proposed document**<br>**206893_11135038_1.0** | **Comments/ Justification for change** |
|---|---|---|
| **Comparison for document: 7776-s260bioal0101 – A144176** | | |
| **1. Summary**<br>**2. Introduction** | **6.2 Initial GNE process versus initial BIP process** | Summarizes the initial comparability study comparing drug product batches from GNE and BIP |
| **3. Material and Methods** | 6.2.1 DP batches tested | Lists the batches tested |
| **4. Product parameter assessed during the comparability study**<br>**5. Results**<br>**6. Assessment of data generated at BI**<br>**7. Other non quantitative assays performed**<br>**8. Conclusion**<br>**9. Analysis performed at Genentech**<br>**10. Assessment of assay** | N/A | The following detailed information is omitted as not considered relevant anymore:<br>• Details on analytical methods valid at the time of the comparability study.<br>• Analysis performed at Genentech<br>• High level flow chart<br>• SDS, IEF and Tryptic map figures<br>Relevant information regarding analysis performed at Boehringer Ingelheim is displayed. Reference is made to Chapter 6.2.2 proposed document. |

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| **Current documents**<br>**7776-s260bioal0101 – A144176**<br>**7776-s260abioal0401 – q00216834-01**<br>**7776-s260abioal0601 – q00259098-01**<br>**206893-10970210_1.0** | **Proposed document**<br>**206893_11135038_1.0** | **Comments/ Justification for change** |
|---|---|---|
| **11. Boehringer Ingelheim Pharma GmbH & Co.KG test results of lots used in clinical trials**<br>**12. Conclusion**<br>**13. Figures** | | |
| **14. Tables**<br>Table 2 – 6 (only test results from Boehringer Ingelheim) | 6.2.2 Results of the analytical comparability study<br>Table 84<br>6.2.3 Conclusion | Summarizes the results of the initial comparability study comparing drug product batches from GNE and BIP<br>Comparability data from drug product batches produced at Boehringer Ingelheim (Table 2 and 3) and Genetech (Table 4 and 5), analysed at Boehringer Ingelheim (Table 6) from the current document are summarized in the proposed document (Table 85). |
| **Comparison for document: 7776-s250abioal0401 – q00216834-01** | | |
| **1. Introduction**<br>**2. Viral filtration process description**<br>**3. Viral removal study**<br>**4. Risk analysis**<br>**5. Revalidation**<br>**6. Assessment** | **6.3 Nanofiltration** | Summarizes the information of the nanofiltration process validation batches. The Viresolve 180 kDa membrane was replaced by Planova BioEx nanofilter(s). |

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| **Current documents**<br><br>**7776-s260bioal0101 – A144176**<br>**7776-s260abioal0401 – q00216834-01**<br>**7776-s260abioal0601 – q00259098-01**<br>**206893-10970210_1.0** | **Proposed document**<br>**206893_11135038_1.0** | **Comments/ Justification for change** |
|---|---|---|
| **7. Summary of data on full scale process validation batches** | | The following detailed information is omitted as not considered relevant anymore or presented elsewhere:<br>• High level flow chart<br>  • Filter characteristics (relevant information for Planova BioEx filter retained in section 3.2.S.2.3 and 3.2.A.2)<br>  • Process description (description for BioEx filter is provided in section 3.2.S.2.2)<br>  • Viral removal study (Reference is made to Section 3.2.A.2)<br>  • Process validation (Reference is made to Section 3.2.S.2.5) |
| **8. Summary of drug substance lot release data of Planova BioEX process validation batches**<br>Table 14 | **6.3.1 Summary of DS release of Planova BioEX nanofiltration process validation batches**<br>Table 86 | Summarizes the results of the nanofiltration process validation batches.<br>The validation data are copied from the current document (Table 14) into the proposed document (Table 86). |

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| **Current documents**<br>**7776-s260bioal0101 – A144176**<br>**7776-s260abioal0401 – q00216834-01**<br>**7776-s260abioal0601 – q00259098-01**<br>**206893-10970210_1.0** | **Proposed document**<br>**206893_11135038_1.0** | **Comments/ Justification for change** |
|---|---|---|
| **Comparison for document: 7776-s260abioal0601 – q00259098-01** | | |
| **1. Comparability-Introduction**<br>**2. Changes to the manufacturing process**<br>**3. Batch information**<br>**4. Reference standards**<br>**5. Test method overview and acceptance criteria** | **6.4 Manganese spiking** | Summarizes the information after implementation of the Mn spike (addition of Manganese to the cell culture process) to improve process robustness including batches tested, reference standard and analytical methods used. |
| **6 Comparability testing results**<br>**6.1 Drug substance release testing**<br>Table 4<br>Figure 1 - 14 | **6.4.1 DS release testing results**<br>Table 87<br>Figure 14 - 27 | Drug substance release testing results are available and copied from Table 4 and Figure 1 – 14 of the current document into the proposed document (Table 87 and Figure 14 – 27).<br>Reference is made to Chapter 6.4.5 of the proposed document regarding additional method description |
| **6.2 Additional analytical characterization**<br>Table 5<br>Figure 15 | **6.4.2 Additional analytical characterization results**<br>Table 88<br>Figure 28 | Additional analytical characterization results are available and copied from Table 5 and Figure 15 of the current document into the proposed document (Table 88 and Figure 28). |

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| **Current documents**<br>**7776-s260bioal0101 – A144176**<br>**7776-s260abioal0401 – q00216834-01**<br>**7776-s260abioal0601 – q00259098-01**<br>**206893-10970210_1.0** | **Proposed document**<br>**206893_11135038_1.0** | **Comments/ Justification for change** |
|---|---|---|
| | | Reference is made to Chapter 6.4.5 of the proposed document regarding additional method description |
| **7. Drug substance stability testing**<br>Table 6 – 8<br>Figure 16 – 18 | **6.4.3 DS stability testing results**<br>Table 89 – 91<br>Figure 29 – 31 | Drug substance stability testing results are available and copied from Table 6 – 8 and Figure 16 - 18 of the current document into the proposed document (Table 89 - 91 and Figure 29 - 31). |
| **8. Conclusion** | **6.4.4 Conclusion** | Provides an overall conclusion |
| **N/A** | **6.4.5 Description of analytical methods** | Short description of the additional analytical methods used |
| **Comparison for document: 206893-10970210_1.0** | | |
| **1. Introduction**<br>**2. Manufacturing process history**<br>**3. Process characterization** | **6.5 Adapted Phenyl Sepharose Hydrophobic Interaction Chromatography** | As part of the lifecycle management, a slight modified (=adapted) purification process of the Phenyl Sepharose chromatography was developed to improve process robustness. |

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| **Current documents**<br>**7776-s260bioal0101 – A144176**<br>**7776-s260abioal0401 – q00216834-01**<br>**7776-s260abioal0601 – q00259098-01**<br>**206893-10970210_1.0** | **Proposed document**<br>**206893_11135038_1.0** | **Comments/ Justification for change** |
|---|---|---|
| | | For manufacturing process history reference is also made to Chapter 3.2.4 of the proposed document.<br>For process characterization reference is made to Chapter 4.3.4.5 and 4.3.4.4 of the proposed document |
| **4. Process comparability studies**<br>**4.1 Outline of comparability testing strategy**<br>**4.2 Overview of Tenecteplase DS batches analsed**<br>**4.3 Acceptance criteria for comparability**<br>**4.4 Drug substance release testing**<br>**4.5 Release testing (side-by-side)**<br>**4.6 Additional characterisation (side-by-side)**<br>**4.7 Removal of process-related impurities**<br>**4.8 DS stability studies** | **6.5.1 Outline of comparability testing strategy**<br>**6.5.2 Overview of Tenecteplase DS batches analysed**<br>**6.5.3 Acceptance criteria for comparability**<br>**6.5.4 Drug substance release testing**<br>**6.5.5 Release testing (side-by-side)**<br>**6.5.6 Additional characterisation (side-by-side)**<br>**6.5.7 Removal of process-related impurities**<br>**6.5.8 DS stability studies** | Comparability results are presented:<br>• Drug substance release testing<br>• Additional characterization<br>• Removal of process related impurities<br>• Drug substance stability studies<br>The complete information of Chapter 4 of the current document is copied into the Chapters 6.5.1 – 6.5.8 of the proposed document. |

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| **Current documents**<br>**7776-s260bioal0101 – A144176**<br>**7776-s260abioal0401 – q00216834-01**<br>**7776-s260abioal0601 – q00259098-01**<br>**206893-10970210_1.0** | **Proposed document**<br>**206893_11135038_1.0** | **Comments/ Justification for change** |
|---|---|---|
| **N/A** | **6.6 Overall analytical comparability assessment** | Overall assessment of the introduced process modifications |
| **N/A** | **7. Conclusion** | Overall conclusion |
| **N/A** | **8. References** | List of references |